What is RAG?

RAG stands for Retrieval-Augmented Generation. It is a technique that combines information retrieval with text generation to improve the accuracy and reliability of large language model responses.

Large language models are trained on data up to a certain cutoff date. This means they do not have knowledge of recent events or private, domain-specific information. RAG solves this problem by retrieving relevant information from an external knowledge source before generating an answer.

The RAG pipeline consists of several stages. First, documents are ingested into the system. Second, these documents are split into smaller chunks. Third, each chunk is converted into a numerical vector using an embedding model. Fourth, these vectors are stored in a vector database such as pgvector. Fifth, when a user asks a question, the system retrieves the most relevant chunks based on vector similarity. Finally, these retrieved chunks are passed to the language model along with the original question to generate a grounded, accurate answer.

RAG reduces hallucination because the model generates answers based on real retrieved text rather than relying purely on its internal parametric memory. It also allows models to access up-to-date and private information without needing to be retrained.